DocuMind AI — Technical Overview & Architecture

1. Executive Summary

DocuMind AI is an enterprise document intelligence and retrieval-augmented generation (RAG) platform.
The application indexes unstructured PDF files and delivers accurate, cited answers via local LLMs.
All inference is strictly executed on the user machine using Ollama and Llama 3.
This local execution guarantees zero data leakage and enterprise-grade privacy for confidential records.
The core pipeline pairs HuggingFace dense embeddings with ChromaDB persistent vector storage.
Every extracted passage maintains strict source metadata, including file names and page indices.

2. Core System Components

- Backend: FastAPI asynchronous web framework with uvicorn server.
- Embeddings Model: BAAI/bge-small-en-v1.5 dense representation (384 dimensions).
- Vector Database: ChromaDB persistent document index.
- Generation Engine: Meta Llama 3 via Ollama Metal-accelerated inference.
- Citation Engine: Exact page-level attribution returned alongside synthesized answers.

DocuMind AI — Security Protocols & Governance

3. Administrative Access & Security Guardrails

DocuMind AI isolates all administrative functions behind an authorization gate.

Upload, listing, and deletion of documents require a valid X-Admin-Key header.

To prevent credentials from being captured in access logs, query parameter keys are prohibited.

Uploaded documents undergo rigorous validation: verification of %PDF magic bytes and size checking.

A maximum upload size of 20 megabytes is enforced to avoid resource exhaustion.

Filenames are sanitized using safe regex replacement to prevent directory traversal exploits.

4. Verification & Grounding Policy

The system enforces a strict anti-hallucination constraint during generation.

When retrieved passages lack sufficient context, the model must explicitly state that the answer is missing.

Answers must never fabricate sources or extrapolate beyond the provided text chunks.